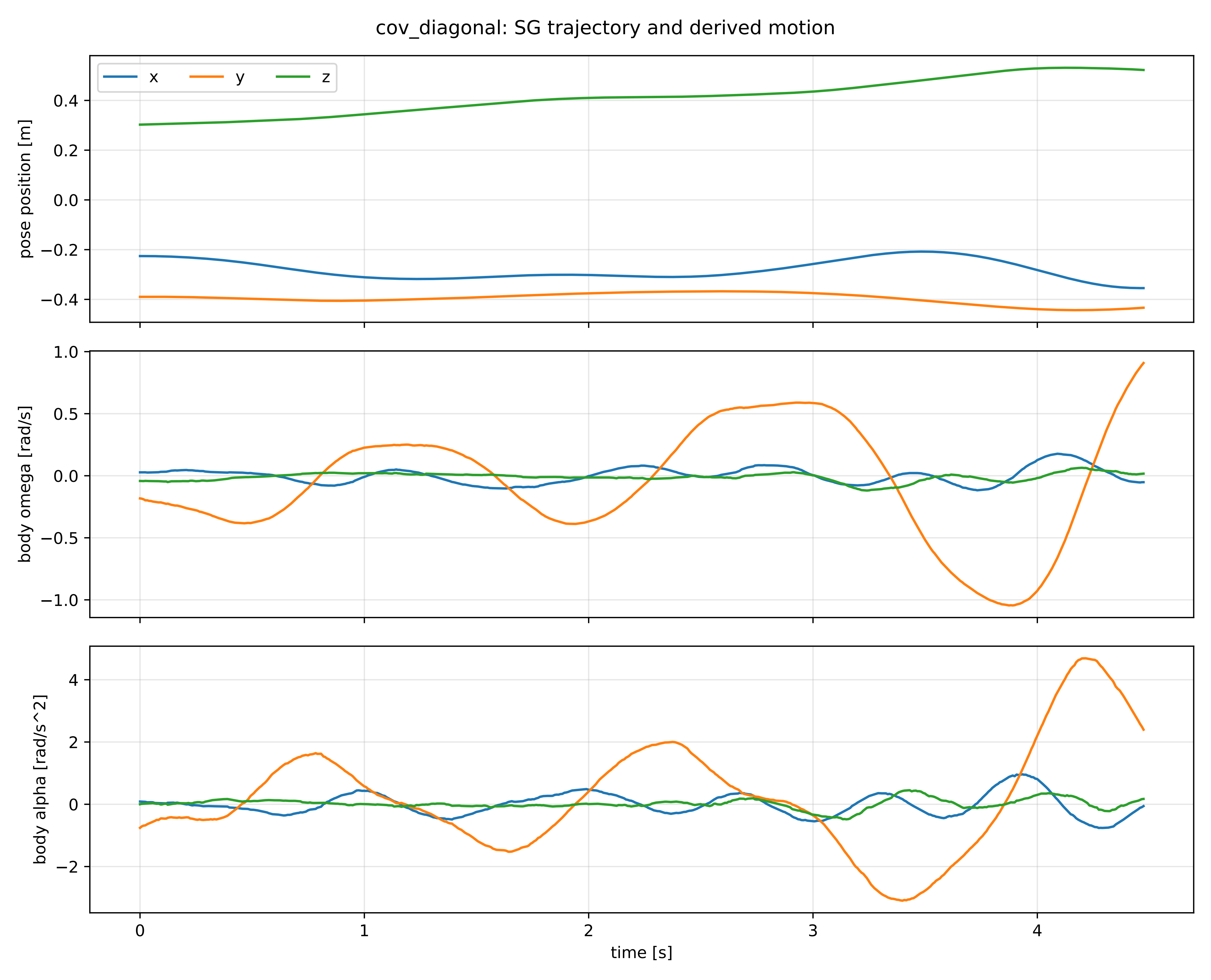
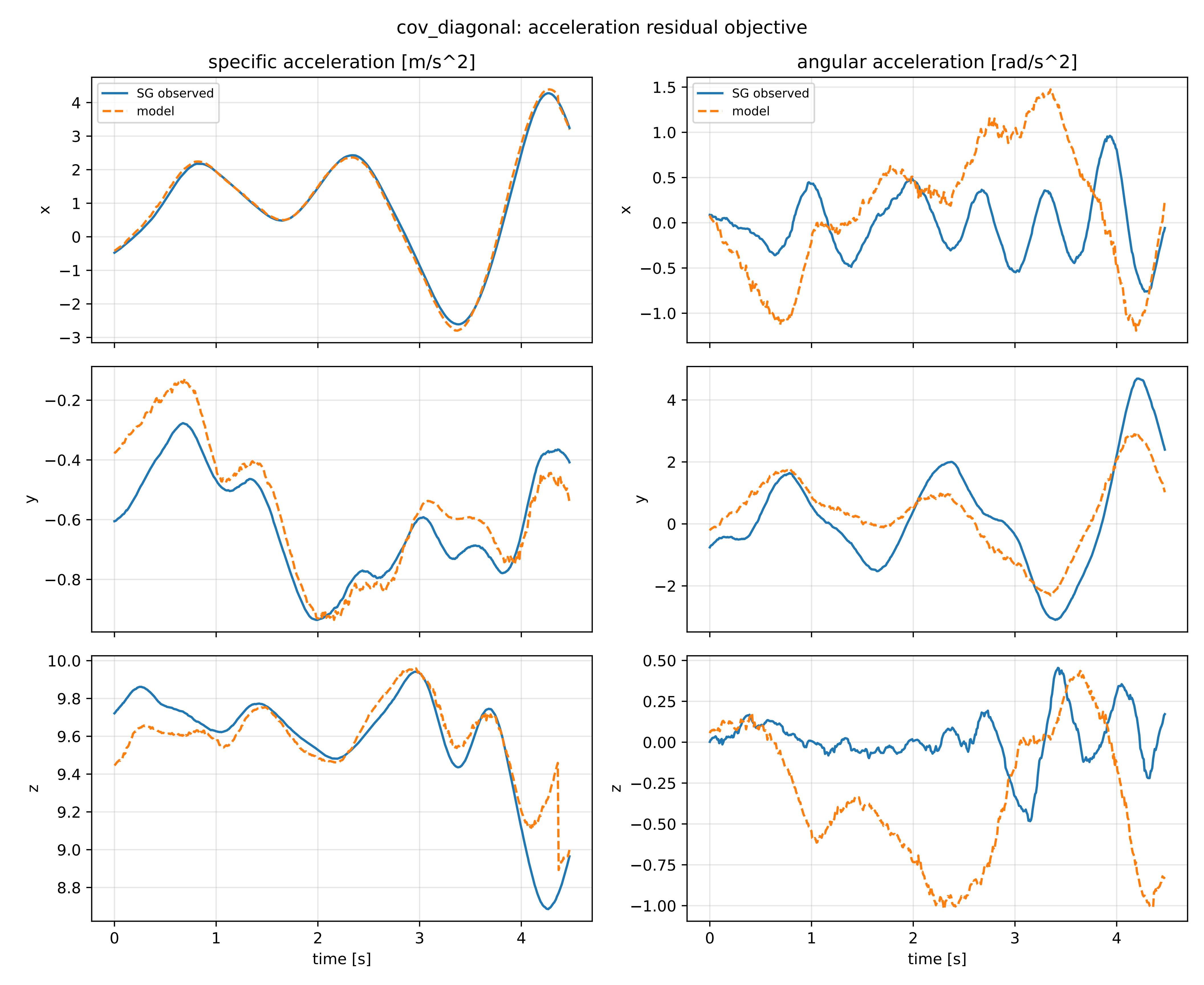
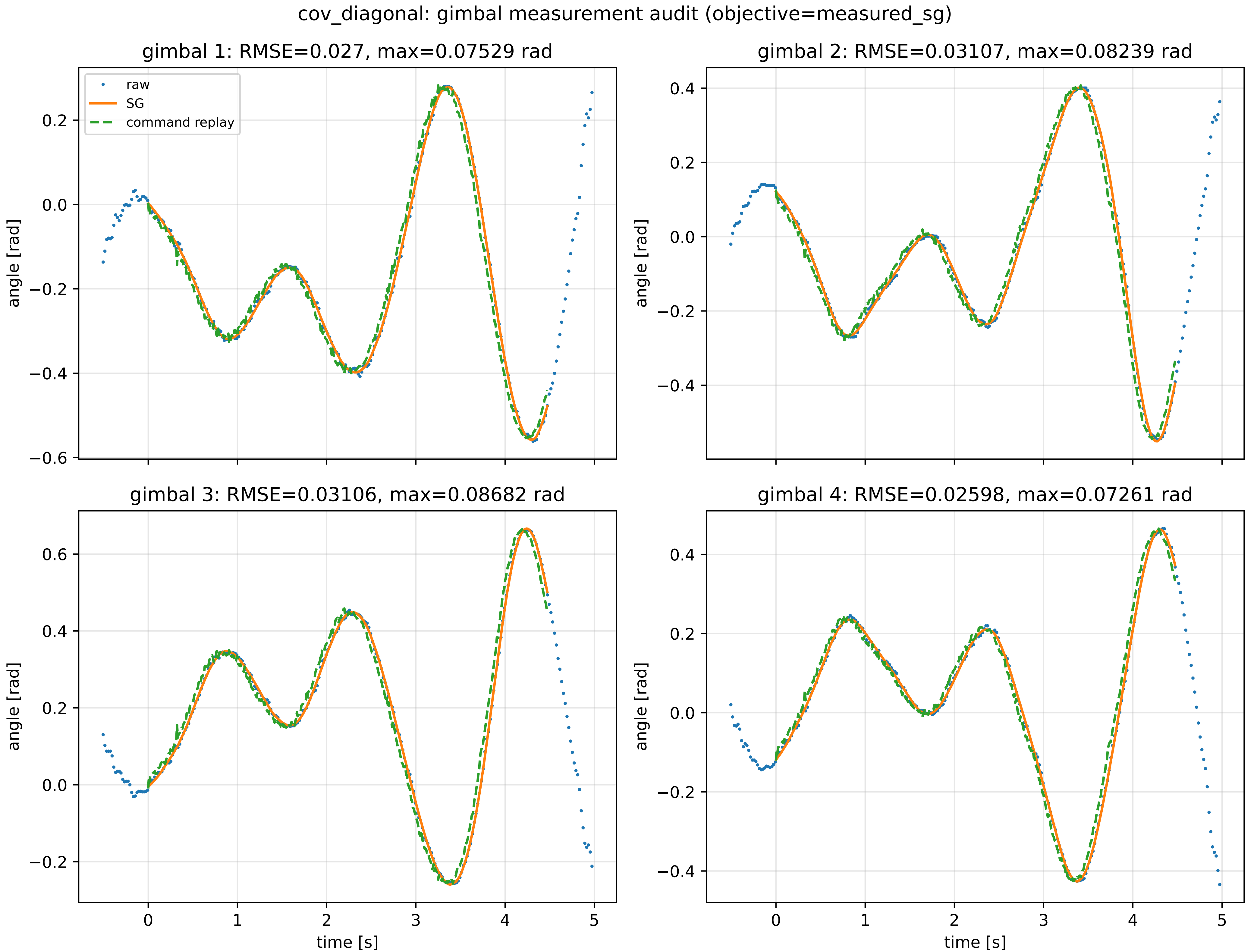
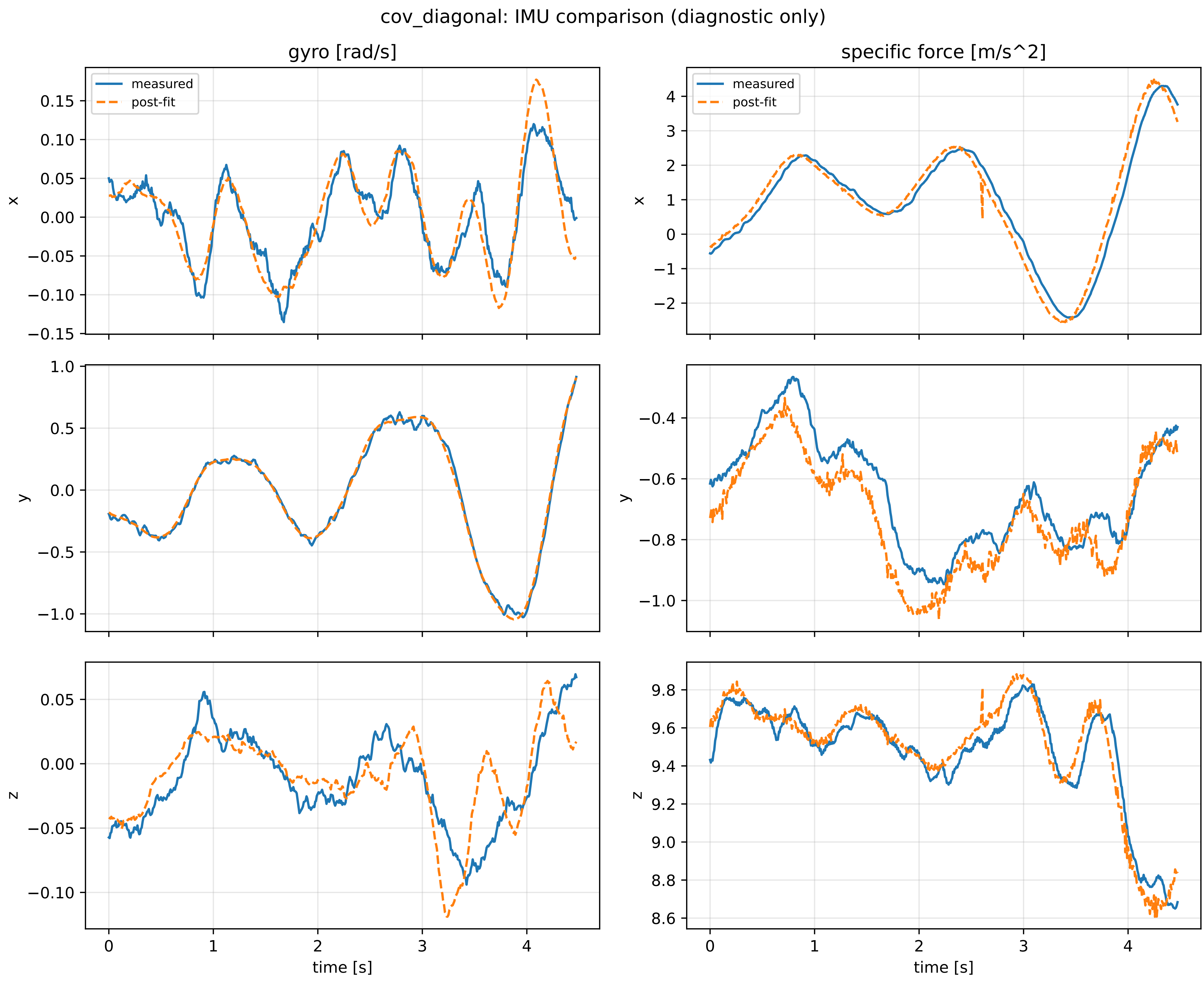


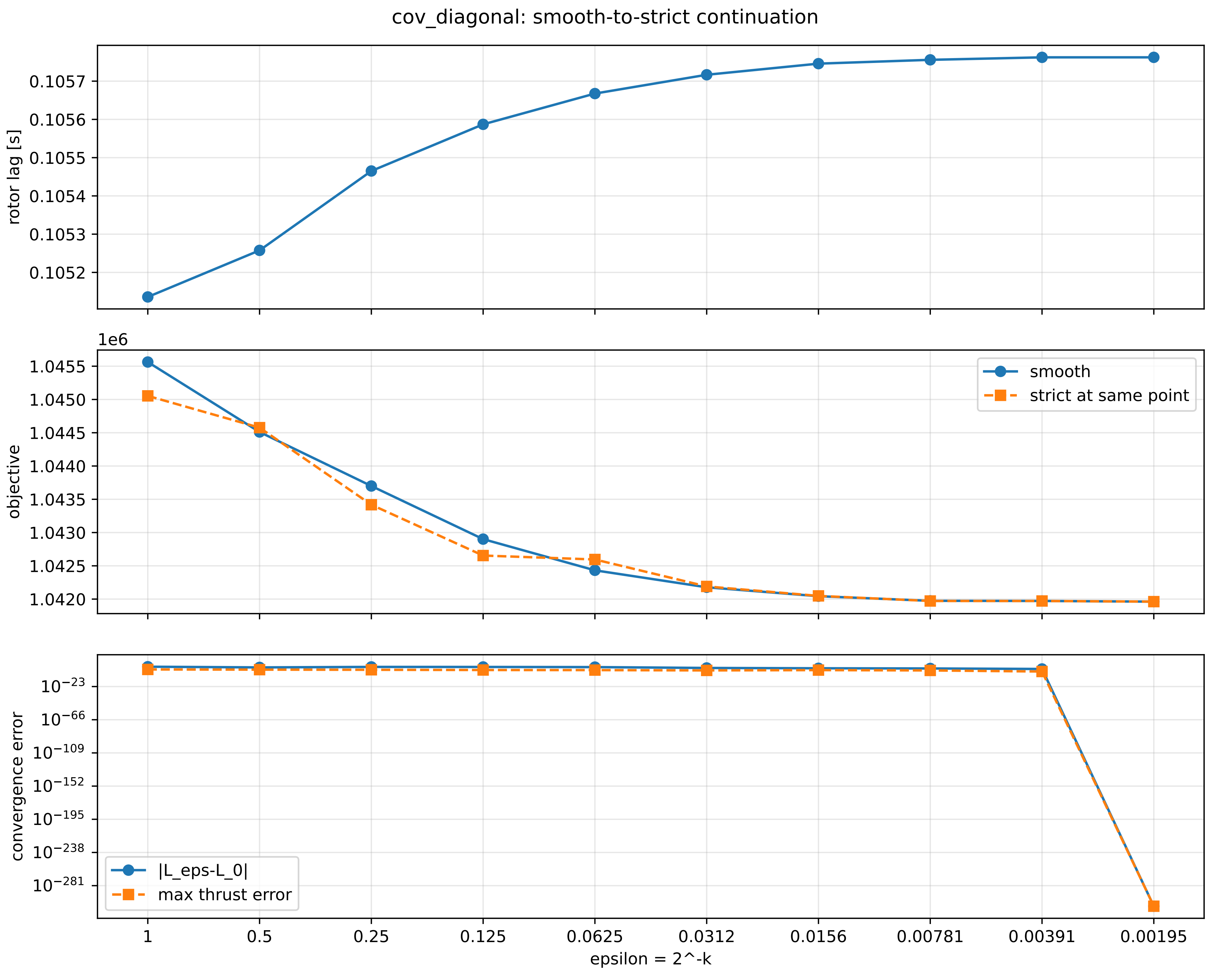
cov_diagonal: SG trajectory and derived motion
x
y
z
pose position [m]
0.4
0.2
0.0
−0.2
−0.4
body omega [rad/s]
1.0
0.5
0.0
−0.5
−1.0
body alpha [rad/s^2]
4
2
0
−2
0
1
2
3
4
time [s]


cov_diagonal: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]



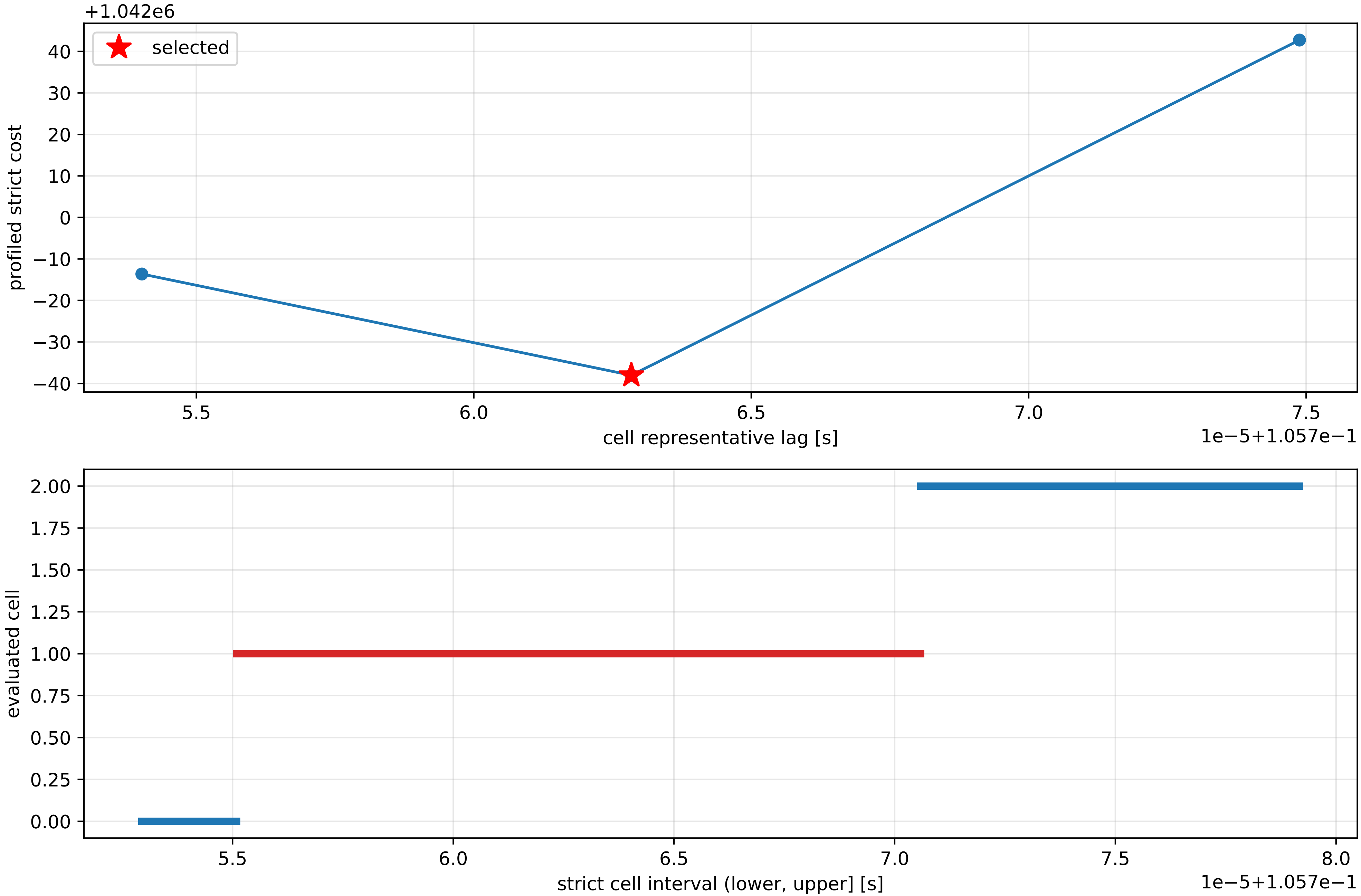
cov_diagonal: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.027, max=0.07529 rad
gimbal 2: RMSE=0.03107, max=0.08239 rad
gimbal 3: RMSE=0.03106, max=0.08682 rad
gimbal 4: RMSE=0.02598, max=0.07261 rad
raw
SG
command replay
angle [rad]
time [s]


cov_diagonal: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



cov_diagonal: smooth-to-strict continuation
rotor lag [s]
0.1057
0.1056
0.1055
0.1054
0.1053
0.1052
1e6
objective
1.0455
1.0450
1.0445
1.0440
1.0435
1.0430
1.0425
1.0420
smooth
strict at same point
convergence error
10^-23
10^-66
10^-109
10^-152
10^-195
10^-238
10^-281
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k

cov_diagonal: exact strict-ZOH lag cells

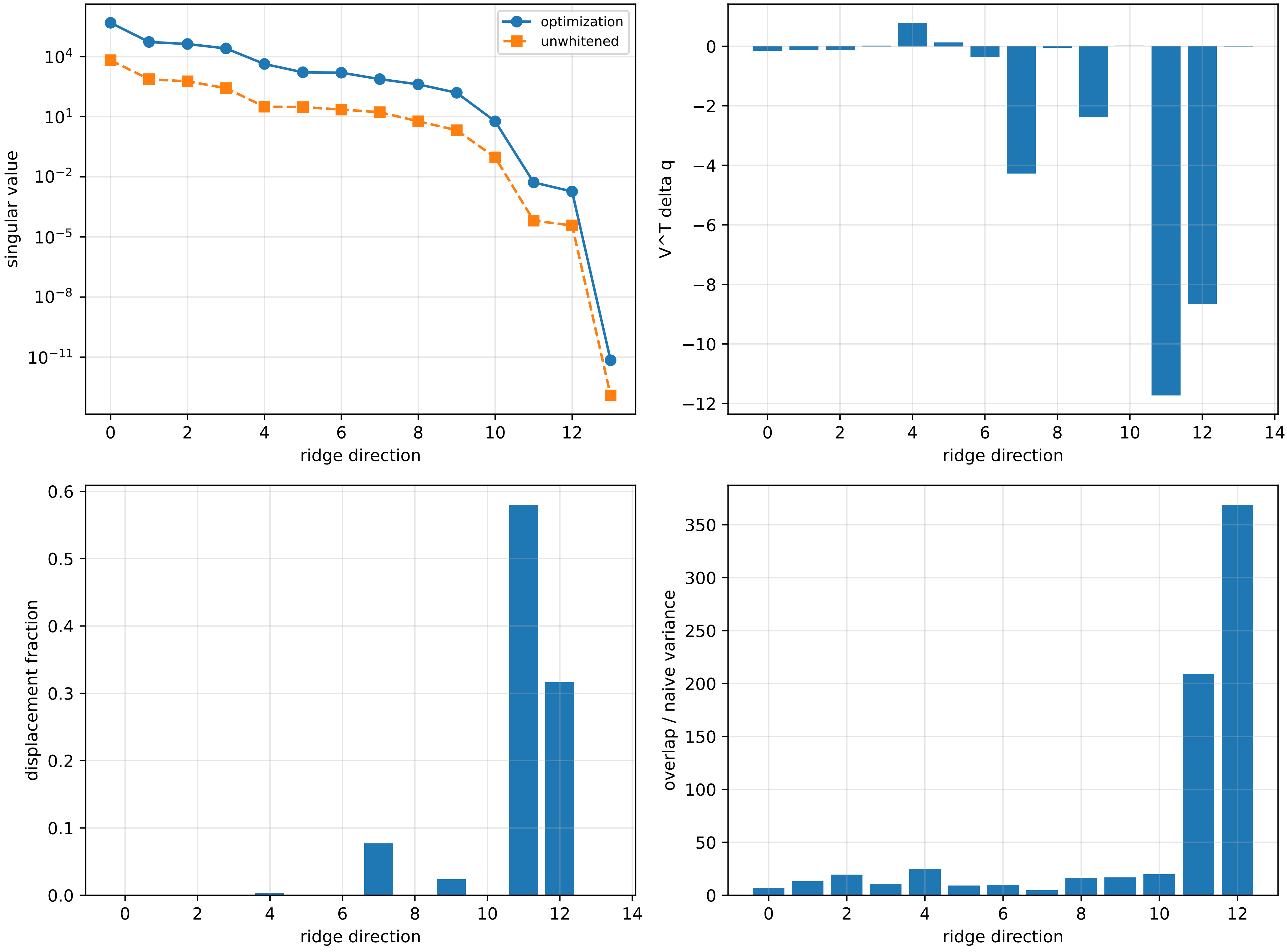


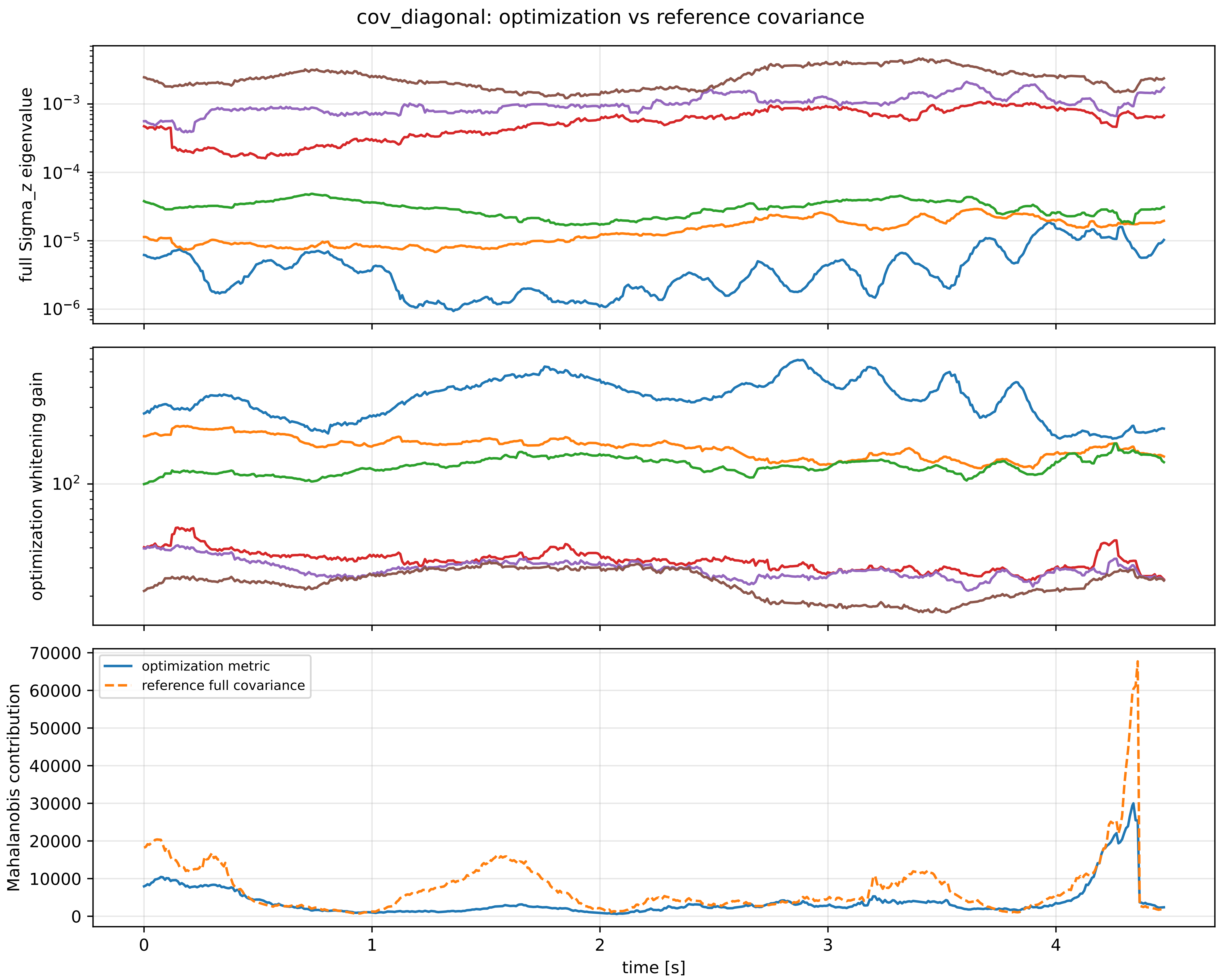
selected (0.105755091, 0.105770588] s; final smooth 0.10576224 s; data boundary=False

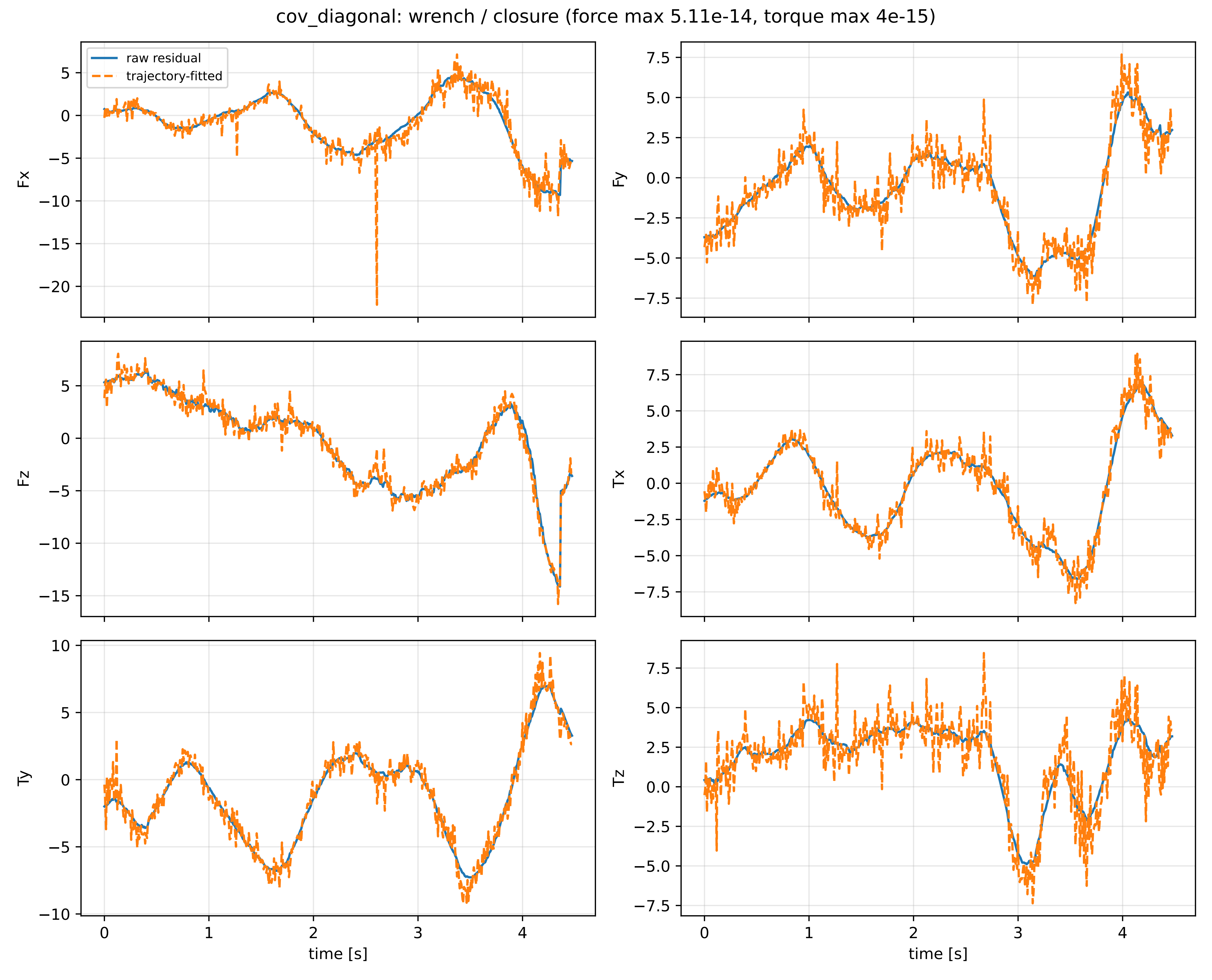
cov_diagonal: parameter estimate

```
Case: cov_diagonal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 16.009702
inertia [kg m^2]:
[[ 2.99723366  2.12369489  1.29148071]
 [ 2.12369489  3.80690974 -1.28854614]
 [ 1.29148071 -1.28854614  5.32025743]]
CoG body [m]: [ 0.08614206  0.15567817 -0.10966165]
force effectiveness: [3.31711631e-05 3.87617103e+00 1.24422437e+01 7.03054481e+00]

scale-free J/m [m^2]:
[[ 0.18721358  0.13265049  0.08066863]
 [ 0.13265049  0.23778767 -0.08048533]
 [ 0.08066863 -0.08048533  0.33231458]]
scale-free f/m: [2.07194132e-06 2.42113878e-01 7.77168976e-01 4.39142765e-01]

rotor lag cell: (0.105755091, 0.105770588] s
representative: 0.105762839 s
```

cov_diagonal: ridge spectrum (rank 13, nullity 1, ||Jv||=1.53e-11)




cov_diagonal: optimization vs reference covariance
full Sigma_z eigenvalue
10^{-3}
10^{-4}
10^{-5}
10^{-6}
optimization whitening gain
10^2
Mahalanobis contribution
70000
60000
50000
40000
30000
20000
10000
0
optimization metric
reference full covariance
0
1
2
3
4
time [s]


cov_diagonal: wrench / closure (force max 5.11e-14, torque max 4e-15)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]